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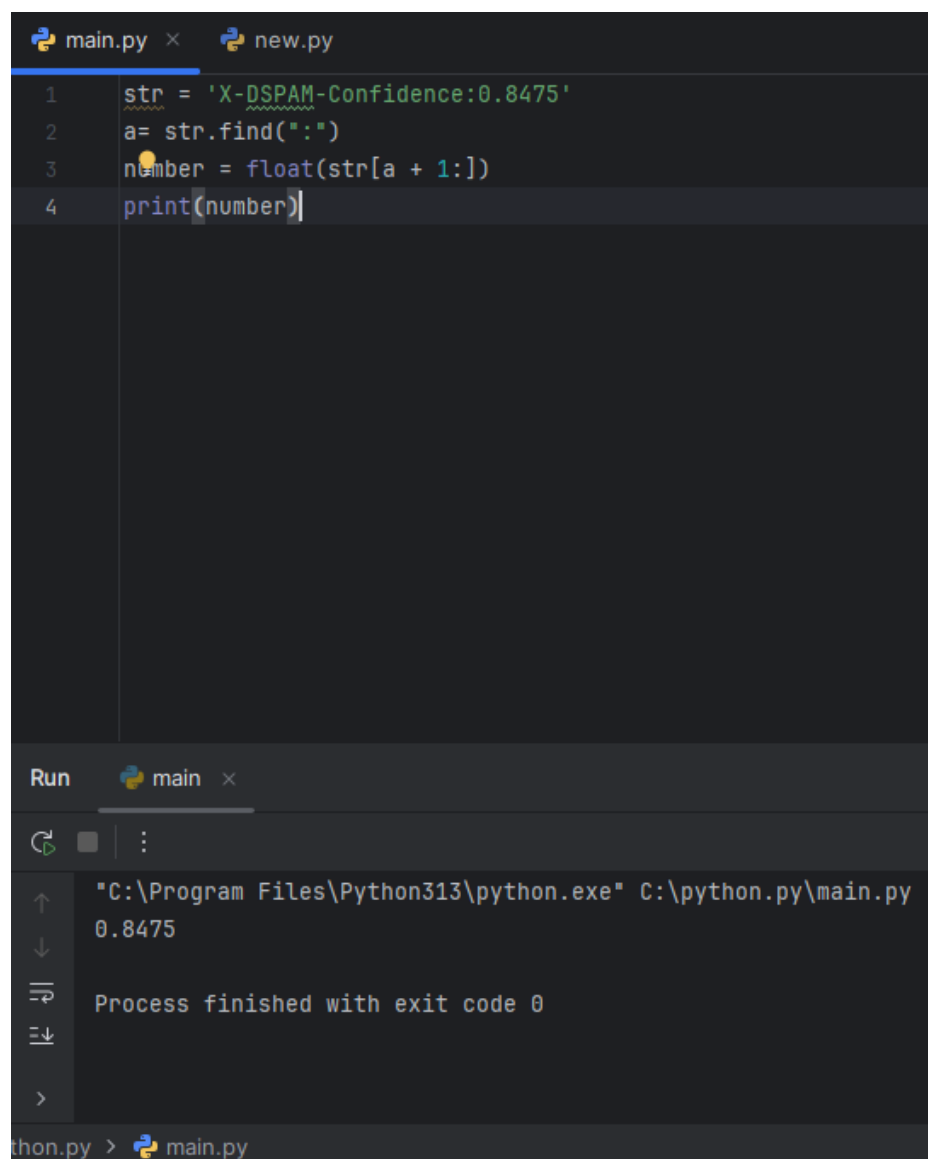
Bài tập chương 6 và 7: Chuỗi và Tệp.

Chương 6: String

Exercise 5: Take the following Python code that stores a string:

```
'str = 'X-DSPAM-Confidence:0.8475'
```

Use find and string slicing to extract the portion of the string after the colon character and then use the float function to convert the extracted string into a floating point number.



The screenshot shows a Python IDE with two tabs: 'main.py' and 'new.py'. The 'main.py' tab is active and contains the following code:

```
1 str = 'X-DSPAM-Confidence:0.8475'
2 a= str.find(":")
3 number = float(str[a + 1:])
4 print(number)
```

Below the code editor, there is a 'Run' button and a 'main' tab. The output console shows the command prompt path and the output of the program:

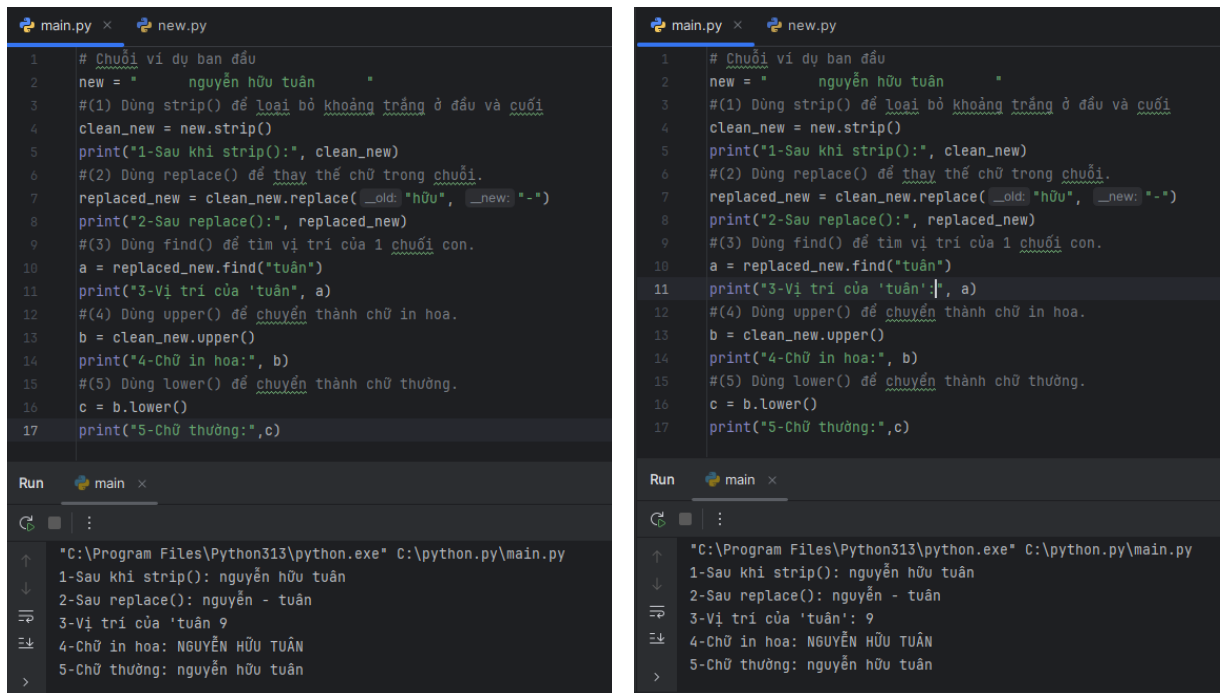
```
"C:\Program Files\Python313\python.exe" C:\python.py\main.py
0.8475
Process finished with exit code 0
```

The status bar at the bottom shows 'thon.py > main.py'.

Exercise 6: Read the documentation of the string methods at <https://docs.python.org/3.5/library/stdtypes.html#string-methods>

You might want to experiment with some of them to make sure you understand how they work. `strip` and `replace` are particularly useful.

The documentation uses a syntax that might be confusing. For example, in `find(sub[, start[, end]])`, the brackets indicate optional arguments. So `sub` is required, but `start` is optional, and if you include `start`, then `end` is optional.



```
1 # Chuỗi ví dụ ban đầu
2 new = "    nguyên hữu tuần    "
3 #(1) Dùng strip() để loại bỏ khoảng trắng ở đầu và cuối
4 clean_new = new.strip()
5 print("1-Sau khi strip():", clean_new)
6 #(2) Dùng replace() để thay thế chữ trong chuỗi.
7 replaced_new = clean_new.replace(_old: "hữu", _new: "-")
8 print("2-Sau replace():", replaced_new)
9 #(3) Dùng find() để tìm vị trí của 1 chuỗi con.
10 a = replaced_new.find("tuần")
11 print("3-Vị trí của 'tuần'", a)
12 #(4) Dùng upper() để chuyển thành chữ in hoa.
13 b = clean_new.upper()
14 print("4-Chữ in hoa:", b)
15 #(5) Dùng lower() để chuyển thành chữ thường.
16 c = b.lower()
17 print("5-Chữ thường:", c)
```

Run main

"C:\Program Files\Python313\python.exe" C:\python.py\main.py
1-Sau khi strip(): nguyên hữu tuần
2-Sau replace(): nguyên - tuần
3-Vị trí của 'tuần': 9
4-Chữ in hoa: NGUYỄN HỮU TUẦN
5-Chữ thường: nguyên hữu tuần

Exercise 1: Write a program to read through a file and print the contents of the file (line by line) all in upper case. Executing the program will look as follows:

python shout.py

Enter a file name: mbox-short.txt

FROM STEPHEN.MARQUARD@UCT.AC.ZA SAT JAN 5 09:14:16 2008

RETURN-PATH: RECEIVED: FROM MURDER (MAIL.UMICH.EDU [141.211.14.90])

BY FRANKENSTEIN.MAIL.UMICH.EDU (CYRUS V2.3.8) WITH LMTPA;

SAT, 05 JAN 2008 09:14:16 -0500

You can download the file from

www.py4e.com/code3/mbox-short.txt

```
main.py x mbox-short.txt
1 # shout.py
2 fname = input("Enter a file name: ")
3 try:
4     fhand = open(fname, 'r', encoding='utf-8')
5 except FileNotFoundError:
6     print(f"File cannot be opened: {fname}")
7     raise SystemExit(1)
8
9 for line in fhand:
10     # loại bỏ ký tự xuống dòng cuối và in chữ hoa
11     print(line.rstrip('\n').upper())

Run main x
RECEIVED: FROM PROD.COLLAB.UHI.AC.UK ([194.35.219.182])
    BY PAPLOO.UHI.AC.UK (JAMES SMTP SERVER 2.1.3) WITH SMTP ID 236
    FOR <SOURCE@COLLAB.SAKAIPROJECT.ORG>;
    THU, 3 JAN 2008 22:18:04 +0000 (GMT)
RECEIVED: FROM NAKAMURA.UITS.IUPUI.EDU (NAKAMURA.UITS.IUPUI.EDU [134.68.220.122])
    BY SHMI.UHI.AC.UK (POSTFIX) WITH ESMTD ID 905D53C2FD
    FOR <SOURCE@COLLAB.SAKAIPROJECT.ORG> THU 3 JAN 2008 22:17:52 +0000 (GMT)
```

Exercise 2: Write a program to prompt for a file name, and then read through the file and look for lines of the form:

X-DSPAM-Confidence:0.8475

When you encounter a line that starts with “X-DSPAM-Confidence:” pull apart the line to extract the floating-point number on the line. Count these lines and then compute the total of the spam confidence values from these lines. When you reach the end of the file, print out the average spam confidence.

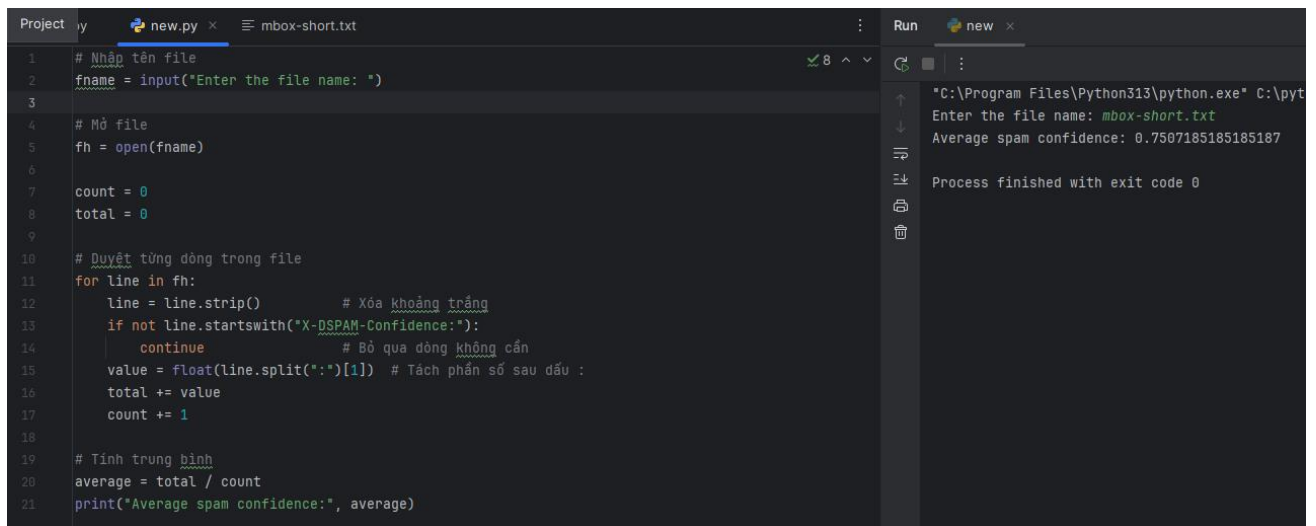
Enter the file name: mbox.txt

Average spam confidence: 0.894128046745

Enter the file name: mbox-short.txt

Average spam confidence: 0.750718518519

Test your file on the mbox.txt and mbox-short.txt files.



```
Project: new.py x mbox-short.txt
1 # Nhập tên file
2 fname = input("Enter the file name: ")
3
4 # Mở file
5 fh = open(fname)
6
7 count = 0
8 total = 0
9
10 # Duyệt từng dòng trong file
11 for line in fh:
12     line = line.strip() # Xóa khoảng trắng
13     if not line.startswith("X-DSPAM-Confidence:"):
14         continue # Bỏ qua dòng không cần
15     value = float(line.split(':')[1]) # Tách phần số sau dấu :
16     total += value
17     count += 1
18
19 # Tính trung bình
20 average = total / count
21 print("Average spam confidence:", average)
```

```
"C:\Program Files\Python313\python.exe" C:\pyt
Enter the file name: mbox-short.txt
Average spam confidence: 0.7507185185185187

Process finished with exit code 0
```

Exercise 3: Sometimes when programmers get bored or want to have a bit of fun, they add a harmless Easter Egg to their program. Modify the program that prompts the user for the file name so that it prints a funny message when the user types in the exact file name “na na boo boo”. The program should behave normally for all other files which exist and don’t exist. Here is a sample execution of the program:

```
python egg.py
```

```
Enter the file name: mbox.txt
```

```
There were 1797 subject lines in mbox.txt
```

```
python egg.py
```

```
Enter the file name: missing.tyxt
```

```
File cannot be opened: missing.tyxt
```

```
python egg.py
```

```
Enter the file name: na na boo boo
```

```
NA NA BOO BOO TO YOU - You have been punk'd!
```

We are not encouraging you to put Easter Eggs in your programs; this is just an exercise.

```
main.py new.py 73.py x mbox-short.txt
1 fname = input("Enter the file name: ")
2
3 # Kiểm tra Easter Egg
4 if fname == "na na boo boo":
5     print("NA NA BOO BOO TO YOU - You have been punk'd!")
6     quit() # Dừng chương trình
7
8 # Thử mở file
9 try:
10     fh = open(fname)
11 except:
12     print("File cannot be opened:", fname)
13     quit()
14
15 count = 0
16 for line in fh:
17     if line.startswith("Subject:"):
18         count += 1
19
20 print("There were", count, "subject lines in", fname)
```

Run 73 x

"C:\Program Files\Python313\python.exe" C:\pyt
Enter the file name: na na boo boo
NA NA BOO BOO TO YOU - You have been punk'd!

Process finished with exit code 0

```
main.py new.py 73.py x mbox-short.txt
1 fname = input("Enter the file name: ")
2
3 # Kiểm tra Easter Egg
4 if fname == "na na boo boo":
5     print("NA NA BOO BOO TO YOU - You have been punk'd!")
6     quit() # Dừng chương trình
7
8 # Thử mở file
9 try:
10     fh = open(fname)
11 except:
12     print("File cannot be opened:", fname)
13     quit()
14
15 count = 0
16 for line in fh:
17     if line.startswith("Subject:"):
18         count += 1
19
20 print("There were", count, "subject lines in", fname)
```

Run 73 x

"C:\Program Files\Python313\python.exe" C:\pyt
Enter the file name: mbox-short.txt
There were 27 subject lines in mbox-short.txt

Process finished with exit code 0

```
main.py new.py 73.py x mbox-short.txt
1 fname = input("Enter the file name: ")
2
3 # Kiểm tra Easter Egg
4 if fname == "na na boo boo":
5     print("NA NA BOO BOO TO YOU - You have been punk'd!")
6     quit() # Dừng chương trình
7
8 # Thử mở file
9 try:
10     fh = open(fname)
11 except:
12     print("File cannot be opened:", fname)
13     quit()
14
15 count = 0
16 for line in fh:
17     if line.startswith("Subject:"):
18         count += 1
19
20 print("There were", count, "subject lines in", fname)
```

Run 73 x

"C:\Program Files\Python313\python.exe"
C:\python.py\73.py
Enter the file name: missing.tyxt
File cannot be opened: missing.tyxt

Process finished with exit code 0